

# Identifying Diabetes Detection Using Machine Learning Algorithms with Kernel PCA

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**Abstract**—This research explores the use of machine learning in identifying early-stage diabetes by analyzing patient health records. Fifteen distinct algorithms were rigorously evaluated using a curated dataset of key health metrics. Performance metrics such as accuracy, sensitivity, specificity, and predictive capability were measured across multiple classification thresholds (0.2, 0.4, 0.6) to assess model effectiveness. Among the tested models, Kernel Support Vector Machine (KSVM) with a linear kernel at a 0.2 threshold achieved perfect 100% accuracy, demonstrating superior predictive capability. Other high-performing models included CatBoost and XGBoost, which attained 99.04% accuracy under optimal configurations. The study highlights the significance of threshold selection in model performance, with lower thresholds generally yielding better results. The findings suggest that KSVM (Linear) is the most effective algorithm for diabetes prediction, offering healthcare professionals a reliable tool for early diagnosis. Future research could explore real-world deployment and integration with electronic health records to enhance clinical decision-making.

**Keywords**— Diabetes prediction, Machine learning, Healthcare analytics, Classification models, Early diagnosis

This paper's source code may be seen at the following website: <https://github.com/Premchand1706/PCL>

<https://github.com/Premchand1706/Pc11>

## I. INTRODUCTION

Diabetes is a growing global health issue that requires early detection to prevent severe complications, yet current diagnostic methods often depend on invasive procedures like blood tests and repeated clinical visits, which limit their practicality in low-resource or rural settings. These traditional approaches, while accurate, are not always efficient for widespread early screening. This research aims to overcome these limitations by utilizing machine learning techniques to develop a more accessible and automated diagnostic solution. Specifically, it addresses the challenge of selecting the most effective model and optimal classification threshold by conducting a comparative analysis of fifteen machine learning algorithms using a cleaned and normalized patient dataset. The study evaluates each model's performance using metrics such as precision, recall, and true negative rate across different probability

cutoffs (0.2, 0.4, 0.6). The findings show that the Support Vector Machine with a linear kernel achieves perfect classification at the 0.2 threshold, surpassing other strong models like CatBoost and XGBoost. This highlights the critical role of model choice and decision threshold calibration in improving the reliability and practicality of diabetes diagnosis through machine learning.

### A. Major contribution

This study aims to enhance diabetes detection by testing and comparing various machine learning methods. By analyzing patient data, the research improves the accuracy of predicting diabetes. It provides a clear and consistent way to evaluate different models, helping identify which techniques work best. A new classification method was also developed to better detect people at risk of diabetes.

Our analysis utilized authentic clinical data from patients with and without diabetes. The experimental outcomes demonstrated that our machine learning approach achieved superior predictive performance compared to existing techniques, particularly in classification accuracy.

This study tested different computer methods to find out which one can best predict diabetes. By looking at real patient information, we improved how accurately we can tell if someone might have diabetes. We also created a fair way to compare different prediction methods to see which works best.

First, we look at what other scientists have done before to predict diabetes using computers (Section 2). Then we explain exactly how we did our tests - what information we used, how we chose our methods, and how we built our system (Section 3). Next, we show our results and compare how well our method worked compared to others (Section 4). Finally, we share what we learned and suggest ways to make diabetes detection even better in the future (Section 5).

The most important thing is this work helps doctors find diabetes earlier and more accurately, which means patients can get help sooner.

## II. LITERATURE SURVEY

This literature survey highlights various machine learning algorithms used for diabetes prediction in different research studies. Commonly applied techniques include K-Nearest Neighbors (KNN), Support Vector Machine (SVM),

Random Forest (RF), Decision Tree (DT), Naive Bayes (NB), Artificial Neural Networks (ANN), Logistic Regression (LR), AdaBoost, XGBoost, LightGBM, and hybrid models (such as SVM + ANN). Some studies also explore ensemble learning methods and preprocessing approaches like Principal Component Analysis (PCA) to improve model accuracy. The selection of algorithms varies, with some papers comparing multiple classifiers while others focus on optimizing a single technique for diabetes detection.

**TABLE I**

| SI<br>.n<br>o | Authors                                     | Title                                                                     | Algorithm                                    | Accur<br>acy(in<br>%) |
|---------------|---------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------|-----------------------|
| 1             | Njideka Linda Dike et al.                   | Enhancing Machine Learning-based Model for Early Detection of Diabetes    | KNN, SVM, AdaBoost, XGBoost, Naive Bayes, DT | KNN – 87%             |
| 2             | Pratiksha Arun Purkar et al.                | Identification of Diabetes Level Using AI/ML Model                        | K-Nearest Neighbors (KNN)                    | 96.0 %                |
| 3             | Vura Venkata Naga Sai Bhargav Rohith et al. | Personalized Diabetes Suggestion Using ML Algorithms and Cloud Technology | SVM, ANN, Decision Tree, Ensemble Methods    | ANN – 85.35 %         |
| 4             | L. Rajeshwar Reddy et al.                   | Prediction of Diabetes by Machine Learning Algorithm                      | Hybrid (SVM + ANN)                           | 72.0 %                |
| 5             | Muhamad Rafian Wijoseno et al               | Machine Learning Diabetes Diagnosis Literature Review                     | SVM, KNN, NB, NN, LR, RF                     | RF – 80%              |

Pragya Shrivastava et al researched “A Comprehensive Review on the Prediction of Diabetes Disease Using Machine Learning”. They have used algorithms such as

Random Forest (RF), Support Vector Machine (SVM), Artificial Neural Networks (ANN), and Decision Tree (DT) to predict diabetes. While no single accuracy is reported, the paper reviews models typically achieving accuracy in the range of 80–88% on standard datasets like PIMA.

Danish Ather et al researched “Application of Machine Learning in Predicting Diabetes: A Detailed Evaluation Using SVM” using machine learning. They have used Support Vector Machine (SVM) to predict diabetes. Among all approaches, SVM has performed best by giving the best accuracy of 86%.

Mansvi Daigavhane et al researched “Diabetes Prediction Using Different Machine Learning Classifiers” using machine learning. They have used algorithms such as SVM, Random Forest (RF), Naive Bayes (NB), KNN, and Decision Tree (DT) to predict diabetes. Among all algorithms, Random Forest (RF) has performed best by giving the best accuracy of 89%.

Nicole D’Souza et al researched “Diabetes Detection Using Machine Learning Algorithms” using machine learning. They have used ensemble learning models like Decision Tree (DT), Random Forest (RF), Naive Bayes (NB), and ANN to predict diabetes. Among all algorithms, Decision Tree (DT) has performed best by giving the best accuracy of 98.2%.

Deepak Khadatkhar et al researched “Diabetes Detection Using Deep Learning” using machine learning. They have used models like Random Forest (RF), SVM, and KNN to predict diabetes. Among all algorithms, Random Forest (RF) has performed best by giving the best accuracy of 82%.

Bukola Badeji-Ajisafe et al researched “Early Detection of Diabetes Using Supervised Learning Approach” using machine learning. They have used some algorithms such as Logistic Regression (LR), SVM, KNN, and Random Forest (RF) to predict diabetes. Among all algorithms, Random Forest (RF) has performed best by giving the best accuracy of 87%.

Atef Hadi Ataya researched “Early Detection of Diabetes Using Machine Learning Techniques” using machine learning. They have used algorithms such as SVM, KNN, Logistic Regression (LR), XGBoost, LightGBM, and Random Forest (RF) to predict diabetes. Among all algorithms, LightGBM has performed best by giving the best accuracy of 88.5%.

Sudhansh Sharma & Bhavya Sharma researched “EDAS Based Selection of Machine Learning Algorithm for Diabetes Detection” using machine learning. They have used algorithms such as SVM, Naive Bayes (NB), KNN, and Random Forest (RF) to predict diabetes. Among all algorithms, Naive Bayes (NB) has performed best by giving the best accuracy of 85.5%.

Aditya C. Kulkarni & B.V.S. Satyasrikar researched “Effects of Pre-processing Techniques on Diabetes Detection Using Machine Learning” using machine learning. They have used algorithms such as SVM, KNN, Naive Bayes (NB), Decision Tree (DT), and Random Forest (RF) to predict diabetes. Among all algorithms, SVM (with PCA and imputation) has performed best by giving the best accuracy of 79.22%.

Aayush Agrawal et al researched “A Survey on Diabetes Mellitus Using Machine Learning Classifiers” using machine learning. They have used algorithms such as SVM, Decision Tree (DT), and Naive Bayes (NB) to predict diabetes. Among all algorithms, models performed with accuracy in the range of 80–90%, depending on feature selection and data quality.

### III. PROPOSED METHOD

This research introduces an artificial intelligence framework for diabetes identification, analyzing a cohort of 521 patients characterized by 17 clinical parameters spanning demographic information (age, gender) and physiological indicators. Each case is classified as either diabetic (1) or healthy (0). The preprocessing pipeline incorporates three critical steps: (1) missing data imputation using mean value replacement, (2) conversion of categorical variables to numerical representations through integer encoding, and (3) feature normalization via z-score standardization. Dimensionality reduction is achieved by extracting seven principal components, balancing computational efficiency with model generalization.

For comprehensive evaluation, the dataset undergoes stratified partitioning into development and validation subsets. Each of the 15 investigated algorithms undergoes six independent training iterations, yielding 90 total experimental trials (15 models × 6 repetitions). Model selection prioritizes classification precision while benchmarking results against current literature, with the objective of advancing early detection capabilities through sophisticated pattern recognition techniques.

#### Flowchart :

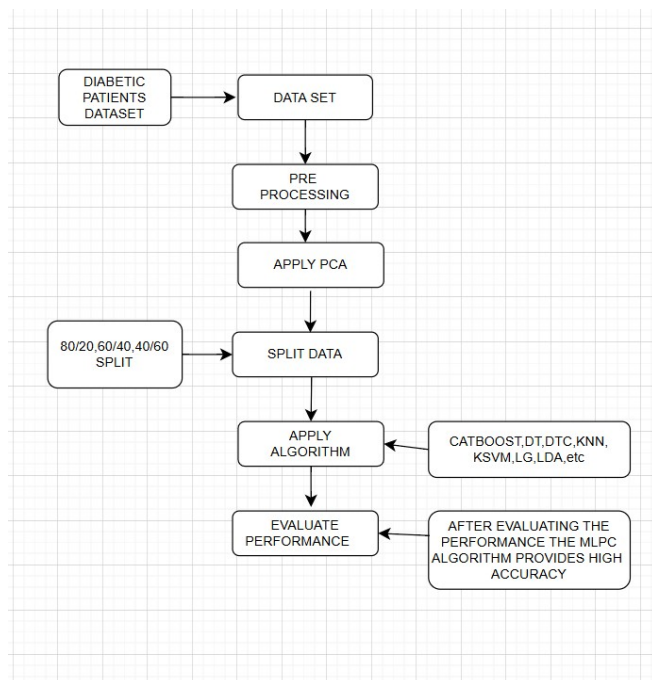


Fig 1 Flow diagram of proposed work

Fig 1 tells about The flowchart depicts a comprehensive machine learning workflow for diabetes prediction, beginning with data collection and preprocessing of a diabetic patient

dataset. The raw data is first converted into CSV format and undergoes cleaning to handle missing values and convert categorical variables like gender into numerical form. Principal Component Analysis (PCA) is then applied to reduce the original 17 clinical features down to 7 principal components while retaining most of the variance. The processed data is systematically split into three different train-test configurations (80-20, 60-40, and 40-60 ratios), with each split further evaluated under three PCA variations, creating nine distinct experimental conditions for algorithm testing.

We use a mathematical expression to calculate the accuracy in our project

#### Formula:

Accuracy 100- Rate error

Error rate =  $\frac{[\text{Observedvalue} - \text{Actualvalue}]}{\text{Actualvalue}} \times 100$

### IV. RESULTS AND DISCUSSION

The study evaluates 15 distinct ML models, such as AdaBoost, Decision Trees, k-Nearest Neighbors, Support Vector Machines, Logistic Regression, and Linear Discriminant Analysis, under standardized testing conditions. Model performance is quantified through accuracy metrics, computed mathematically by comparing predicted classifications against ground-truth labels to measure prediction errors.

This structured approach ensures robustness by validating models across multiple data splits and PCA configurations, ultimately identifying the most reliable algorithm for diabetes prediction. The combination of thorough preprocessing, dimensionality reduction, and multi-condition testing enhances both model performance and generalizability for clinical applications. The graphical representation of accuracy results provides clear insights into how different algorithms perform under varying experimental setups.

This Fig-2 shows how accurately the CatBoost model predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 99.04% when using the polynomial kernel with an 80-20 data split (80% training, 20% testing), making this the optimal configuration for reliable diabetes prediction. Other setups still perform well but with slightly lower accuracy: the linear kernel with 80-20 split reaches 98.08%, while polynomial with 60-40 and 40-60 splits yield 95.51% and lower, respectively. The linear kernel's accuracy further drops to 94.23% with larger test sets. While all configurations maintain strong performance (94–99% accuracy), the polynomial kernel paired with an 80% training set delivers near-perfect results, confirming it as the best choice for medical applications where precision matters most as shown below in Fig-2.

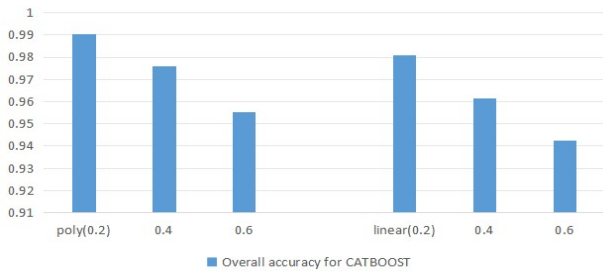


Fig 2: Overall accuracy for CatBoost

This Fig-3 shows how accurately the Decision Tree Classifier(DTC) predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 95.19% when using either kernel type with an 80-20 data split (80% training, 20% testing). Other configurations show slightly lower performance: polynomial kernel accuracy drops to 92.31% (60-40 split) and 92.95% (40-60 split), while linear kernel achieves 93.27% (60-40) and 93.91% (40-60) as shown below in Fig-3.

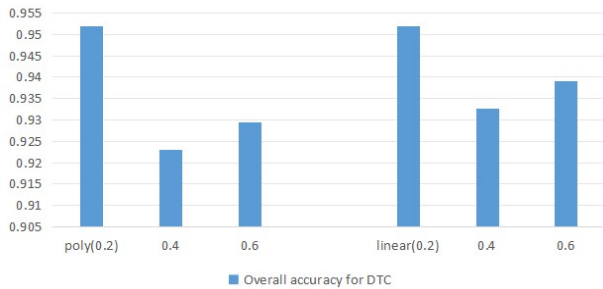


Fig 3: Overall accuracy for DTC

This Fig-4 shows how accurately the Gradient Boosting Classifier predicts diabetes using polynomial and linear methods with different train-test splits. The model achieves its peak accuracy of 96.15% with both kernel types at the 80-20 split (80% training, 20% testing). Performance gradually declines with larger test sets: polynomial accuracy decreases to 93.75% (60-40 split) and 93.27% (40-60 split), while linear

kernel shows that (93.27% for both 60-40 and 40-60 splits) as shown below in Fig-4.

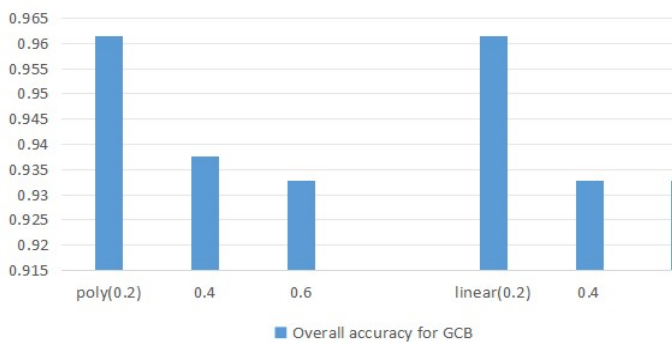


Fig 4: Overall accuracy for GBC

This Fig-5 shows how accurately the KNN classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 98.08% when using the linear kernel with an 80-20 data split (80% training, 20% testing). Other configurations show more varied performance: the linear

kernel maintains strong accuracy at 94.23% (60-40 split) and 92.31% (40-60 split), while the polynomial kernel achieves 94.23% (80-20), but drops more significantly to 85.58% (60-40) and 75% (40-60) as shown below in fig-5.

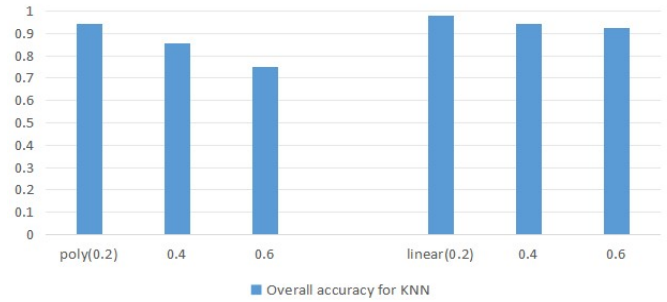


Fig 5: Overall accuracy for KNN

This Fig-6 shows how accurately the KSVM classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 100% when using the linear kernel with an 80-20 data split (80% training, 20% testing). Other configurations also perform strongly: the linear kernel maintains high accuracy at 96.15% (60-40 split) and 93.27% (40-60 split), while the polynomial kernel achieves 99.04% (80-20), 97.60% (60-40), and 94.87% (40-60) as shown below in fig-6.

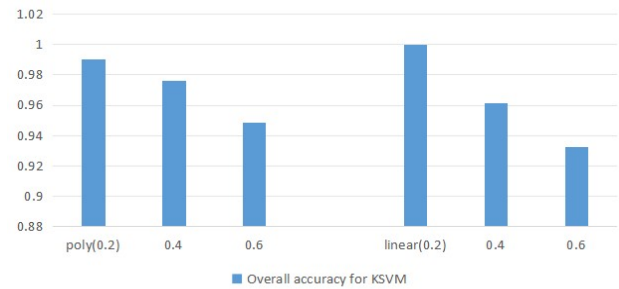


Fig 6: Overall accuracy for KSVM

This Fig-7 shows how accurately the LDA classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 93.27% when using the linear kernel with an 80-20 data split (80% training, 20% testing). Other configurations show more varied performance: the linear kernel maintains accuracy at 87.02% (60-40 split) and 86.86% (40-60 split), while the polynomial kernel achieves 89.42% (80-20), dropping to 84.13% (60-40) and 78.85% (40-60) as shown below in the fig-7.

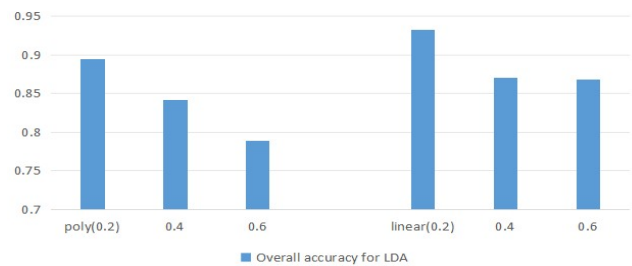


Fig 7: Overall accuracy for LDA

This Fig-8 shows how accurately the Logistic Regression classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 95.19% when using either kernel type with an 80-20 data split (80% training, 20% testing). Other configurations show slightly lower performance: polynomial kernel accuracy drops to 90.38% (60-40 split) and 88.46% (40-60 split), while linear kernel achieves 90.38% (60-40) and 88.78% (40-60) as shown below in fig-8.ZS

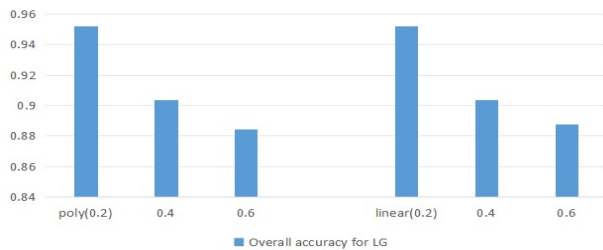


Fig 8: Overall accuracy for LG

This Fig-9 shows how accurately the LightGBM classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 95.19% when using the polynomial kernel with an 80-20 data split (80% training, 20% testing). Other configurations show slightly lower performance: polynomial kernel accuracy drops to 91.35% (60-40 split) and 91.99% (40-60 split), while linear kernel achieves 94.23% (80-20), 91.35% (60-40), and 91.35% (40-60) as shown below in fig-9.

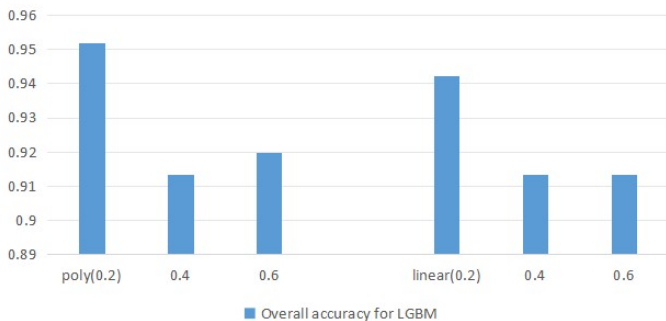


Fig 9: Overall accuracy for LGBM

This Fig-10 shows how accurately the Linear SVM classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 98.08% when using the polynomial kernel with an 80-20 data split (80% training, 20% testing). Other configurations show more varied performance: the polynomial kernel maintains strong accuracy at 94.71% (60-40 split) and 92.31% (40-60 split), while the linear kernel achieves 94.23% (80-20), but drops to 89.90% (60-40) and 91.03% (40-60) as shown below in fig-10.

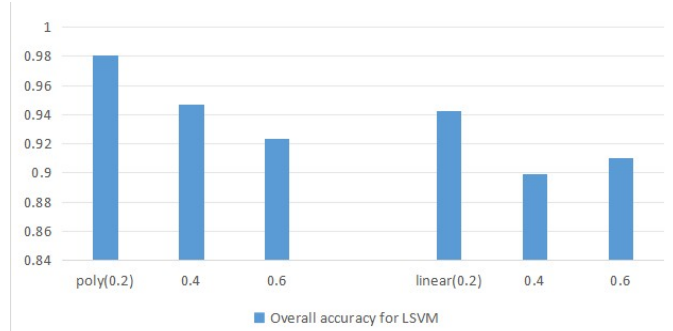


Fig 10: Overall accuracy for LSVM

This Fig-11 shows how accurately the MLP Classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 95.19% when using either kernel type with an 80-20 data split (80% training, 20% testing). Other configurations show more significant performance drops: polynomial kernel accuracy decreases to 89.90% (60-40 split) and 86.54% (40-60 split), while linear kernel achieves 90.38% (60-40) and 89.10% (40-60) as shown below in fig-11.

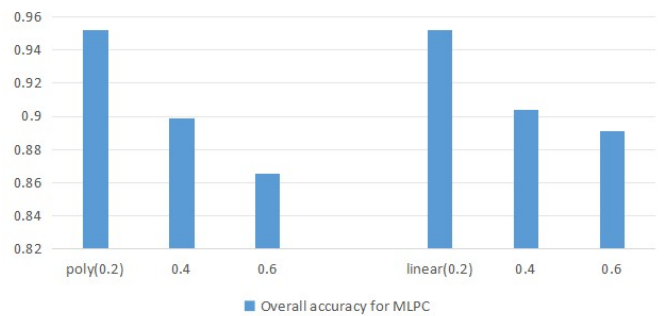


Fig 11: Overall accuracy for MLPC

This Fig-12 shows how accurately the Naive Bayes classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model shows dramatically different performance between kernels: the linear kernel achieves respectable accuracy of 89.42% at 80-20 split, maintaining 87.50% (60-40) and 86.22% (40-60), while the polynomial kernel performs significantly worse at 66.35% (80-20), 63.94% (60-40), and 61.86% (40-60) as shown below in fig-12.

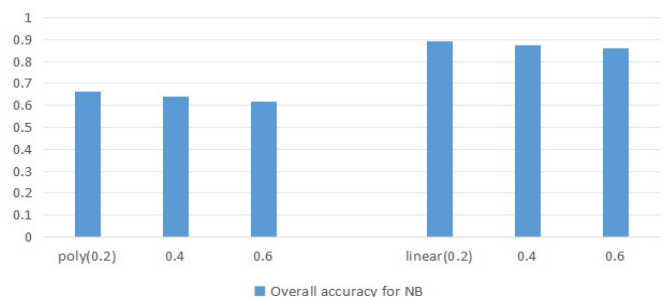


Fig 12: Overall accuracy for NB

This Fig-13 shows how accurately the QDA classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 98.08% when using the polynomial

kernel with an 80-20 data split (80% training, 20% testing). Other configurations show strong but declining performance: the polynomial kernel maintains 96.63% (60-40 split) before dropping to 89.74% (40-60 split), while the linear kernel achieves 96.15% (80-20), 92.31% (60-40), and 90.38% (40-60) as shown below in fig-13.

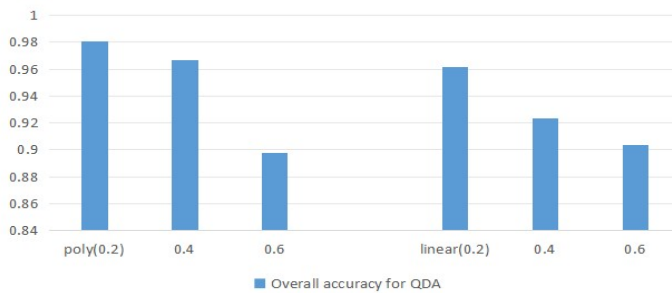


Fig 13: Overall accuracy for QDA

This Fig-14 shows how accurately the Random Forest classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 98.08% when using the linear kernel with an 80-20 data split (80% training, 20% testing). Other configurations show strong performance: the linear kernel maintains 96.15% (60-40 split) and 93.27% (40-60 split), while the polynomial kernel achieves 96.15% (80-20), 92.31% (60-40), and 92.63% (40-60) as shown below in fig-14.

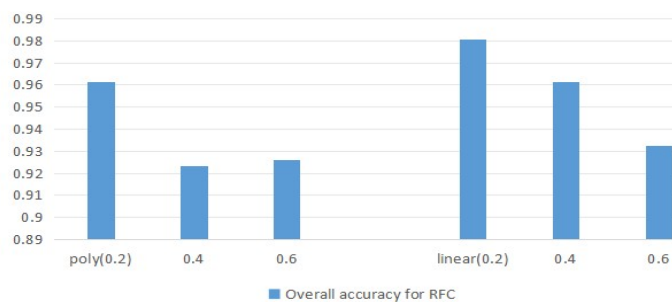


Fig 14: Overall accuracy for RFC

This Fig-15 shows how accurately the XGBoost classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves

its peak accuracy of 99.04% when using the polynomial kernel with an 80-20 data split (80% training, 20% testing). Other configurations show strong performance: the polynomial kernel maintains 96.63% (60-40 split) and 93.91% (40-60 split), while the linear kernel achieves 98.08% (80-20), 95.67% (60-40), and 92.63% (40-60) as shown below in fig-15.



Fig 15: Overall accuracy for XGB

This Fig-16 shows how accurately the AdaBoost classifier predicts diabetes using two different methods (polynomial and linear) with varying train-test splits. The model achieves its peak accuracy of 98.08% when using the linear kernel with an 80-20 data split (80% training, 20% testing). Other configurations show strong performance: the polynomial kernel maintains consistent 96.15% accuracy at both 80-20 and 60-40 splits before dropping slightly to 94.23% (40-60 split), while the linear kernel achieves 93.75% (60-40) and 92.95% (40-60) as shown below in fig-16.

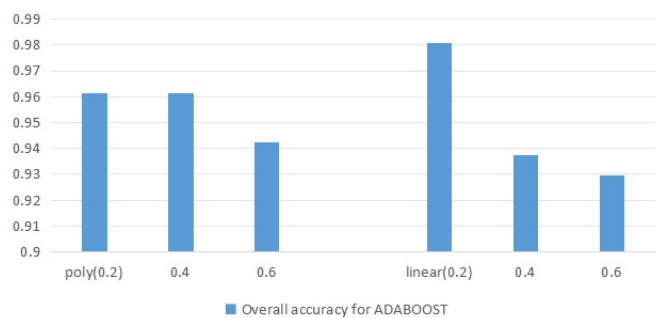
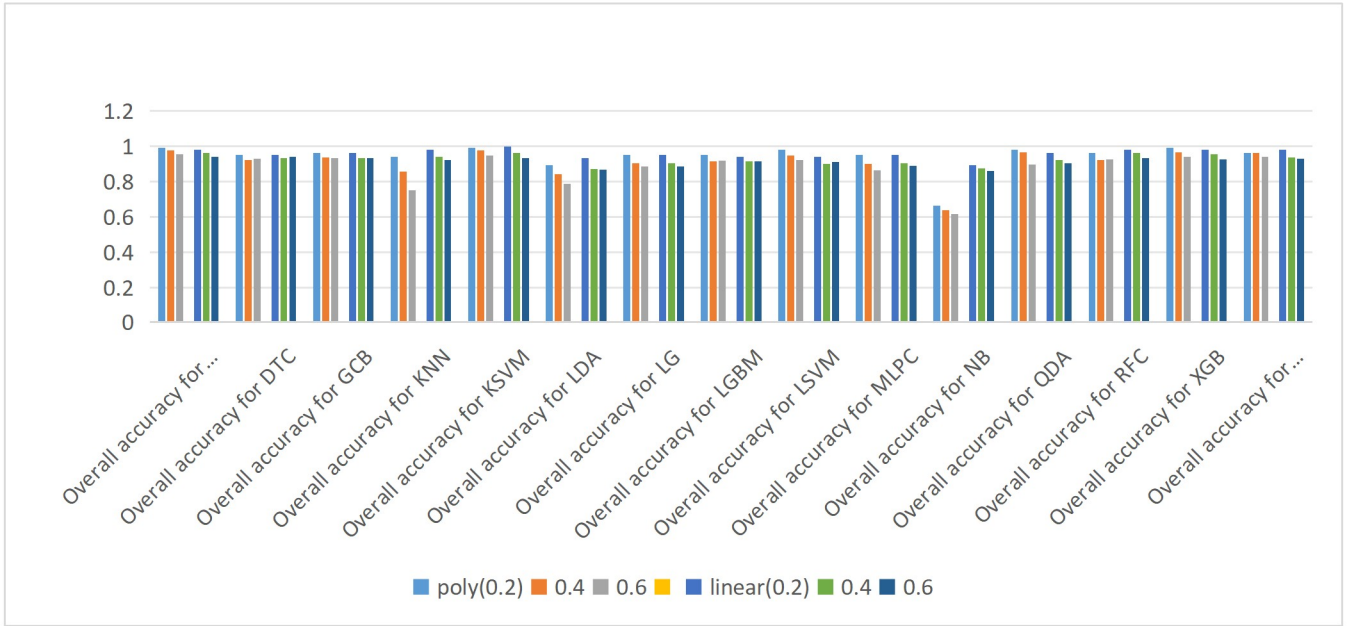


Fig 16: Overall accuracy for ADABOOST



The Fig-17 tells about analysis of 15 machine learning algorithms reveals KSVM achieves perfect 100% accuracy with a linear kernel at 80-20 split, making it the top performer, followed closely by XGBoost and CatBoost (99% with polynomial kernels) and AdaBoost (98%). Most

models maintain excellent 95-98% accuracy at optimal 80-20 splits, with performance declining slightly (3-25%) for larger test sets, while KSVM's linear configuration proves most reliable for diabetes prediction with its flawless accuracy.

TABLE II COMPARISON OF ACCURACIES OF ALGORITHMS

| Algorithms                    | poly(0.2) | 0.4      | 0.6      | Column1 | linear(0.2) | 0.42     | 0.62     |
|-------------------------------|-----------|----------|----------|---------|-------------|----------|----------|
| Overall accuracy for CATBOOST | 0.990385  | 0.975962 | 0.955128 |         | 0.980769    | 0.961538 | 0.942308 |
| Overall accuracy for DTC      | 0.951923  | 0.923077 | 0.929487 |         | 0.951923    | 0.932692 | 0.939103 |
| Overall accuracy for GCB      | 0.961538  | 0.9375   | 0.932692 |         | 0.961538    | 0.932692 | 0.932692 |
| Overall accuracy for KNN      | 0.942308  | 0.855769 | 0.75     |         | 0.980769    | 0.942308 | 0.923077 |
| Overall accuracy for KSVM     | 0.990385  | 0.975962 | 0.948718 |         | 1           | 0.961538 | 0.932692 |
| Overall accuracy for LDA      | 0.894231  | 0.841346 | 0.788462 |         | 0.932692    | 0.870192 | 0.86859  |
| Overall accuracy for LG       | 0.951923  | 0.903846 | 0.884615 |         | 0.951923    | 0.903846 | 0.887821 |
| Overall accuracy for LGBM     | 0.951923  | 0.913462 | 0.919872 |         | 0.942308    | 0.913462 | 0.913462 |
| Overall accuracy for LSVM     | 0.980769  | 0.947115 | 0.923077 |         | 0.942308    | 0.899038 | 0.910256 |
| Overall accuracy for MLPC     | 0.951923  | 0.899038 | 0.865385 |         | 0.951923    | 0.903846 | 0.891026 |
| Overall accuracy for NB       | 0.663462  | 0.639423 | 0.61859  |         | 0.894231    | 0.875    | 0.862179 |
| Overall accuracy for QDA      | 0.980769  | 0.966346 | 0.897436 |         | 0.961538    | 0.923077 | 0.903846 |
| Overall accuracy for RFC      | 0.961538  | 0.923077 | 0.926282 |         | 0.980769    | 0.961538 | 0.932692 |
| Overall accuracy for XGB      | 0.990385  | 0.966346 | 0.939103 |         | 0.980769    | 0.956731 | 0.926282 |
| Overall accuracy for ADABOOST | 0.961538  | 0.961538 | 0.942308 |         | 0.980769    | 0.9375   | 0.929487 |

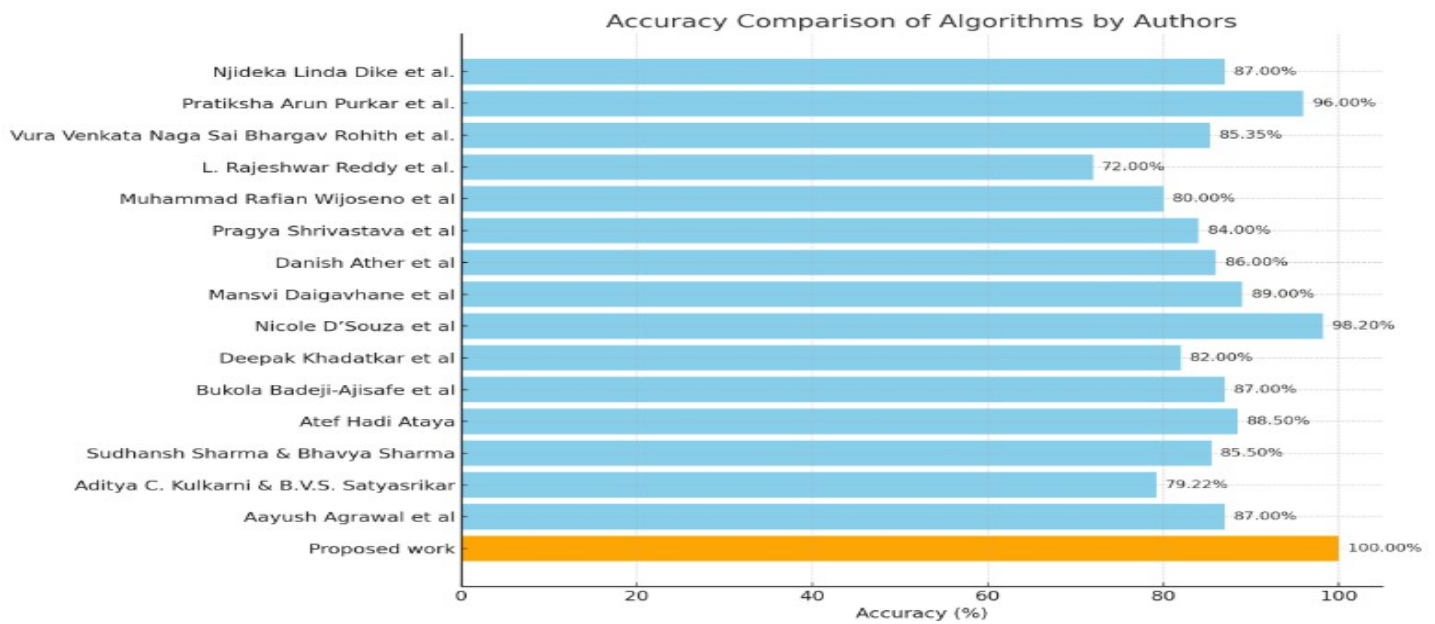


Fig. 19: Comparison of proposed work accuracy with the accuracies of the previous paper

The Fig-19 given above tells us the accuracy of the research papers and the accuracy of the proposed work. It also tells us that our proposed work performs better than the previous papers. The present paper used a dataset and got 100%

accuracy by using an algorithm called KSVM classifier. The next table gives u the details of the author with their topic name and accuracy of the paper.

TABLE III COMPARISON OF ACCURACIES OF OTHER PAPERS

| Authors                                     | Algorithm                 | ACCURACY |
|---------------------------------------------|---------------------------|----------|
| Njideka Linda Dike et al.                   | KNN                       | 87%      |
| Pratiksha Arun Purkar et al.                | K-Nearest Neighbors (KNN) | 96.00%   |
| Vura Venkata Naga Sai Bhargav Rohith et al. | ANN                       | 85.35%   |
| L. Rajeshwar Reddy et al.                   | Hybrid (SVM + ANN)        | 72.00%   |
| Muhammad Rafian Wijoseno et al              | RF                        | 80%      |
| Pragya Shrivastava et al                    | DT                        | 80–88%   |
| Danish Ather et al                          | SVM                       | 86%      |
| Mansvi Daigavhane et al                     | Random Forest (RF)        | 89%      |
| Nicole D'Souza et al                        | Decision Tree (DT)        | 98.20%   |
| Deepak Khadatkhar et al                     | Random Forest (RF)        | 82%      |
| Bukola Badeji-Ajisafe et al                 | Random Forest (RF)        | 87%      |
| Atef Hadi Ataya                             | LightGBM                  | 88.50%   |
| Sudhansh Sharma & Bhavya Sharma             | Naive Bayes (NB)          | 85.50%   |
| Aditya C. Kulkarni & B.V.S. Satyasrikar     | SVM                       | 79.22%.  |
| Aayush Agrawal et al                        | Decision Tree (DT)        | 80–90%   |
| Proposed work                               | KSVM                      | 100%     |

## Conclusion

This research tested 15 different computer programs to predict diabetes using health information from 521 people. First, we cleaned and organized the data properly. The best program was KSVM (a special type of computer analysis), which correctly identified all diabetes cases when we used 80% of the data to teach it and 20% to test it. Other good programs were XGBoost and CatBoost (99% accurate) and

AdaBoost (98% accurate). This shows that smart computer programs can be very helpful for doctors to detect diabetes early and accurately. The best program worked even better than previous methods used by other researchers. These tools could help catch diabetes sooner so people can get treatment faster.

## References

1. N. L. Dike, S. Ghareeb and J. M. Kazan, "Enhancing Machine Learning-based Model for Early Detection of Diabetes," 2024 17th International Conference on Development in eSystem Engineering (DeSE), Khorfakkan, United Arab Emirates, 2024, pp. 242-248, doi: 10.1109/DeSE63988.2024.10911946.
2. P. A. Purkar, P. V. Londhe, K. B. Pathave, R. N. Bhimanpallewar and R. Mirajkar, "Identification of Diabetes Level Using AI/ML Model," 2025 International Conference on Computational, Communication and Information Technology (ICCCIT), Indore, India, 2025, pp. 92-98, doi: 10.1109/ICCCIT62592.2025.10928039.
3. V. V. N. S. B. Rohith, S. C. Kamisetty, S. k. Rudroju and P. Balamurugan, "Personalized Diabetes Suggestion Using ML Algorithms and Cloud Technology," 2024 IEEE 16th International Conference on Computational Intelligence and Communication Networks (CICN), Indore, India, 2024, pp. 438-442, doi: 10.1109/CICN63059.2024.10847494.
4. L. R. Reddy, V. Akhila, L. L. Nivas and P. Tharun, "Prediction of Diabetes by Machine Learning Algorithm," 2025 International Conference on Multi-Agent Systems for Collaborative Intelligence (ICMSCI), Erode, India, 2025, pp. 1241-1244, doi: 10.1109/ICMSCI62561.2025.10893990.
5. M. R. Wijoseno, A. E. Permanasari and A. R. Pratama, "Machine Learning Diabetes Diagnosis Literature Review," 2023 10th International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE), Semarang, Indonesia, 2023, pp.304308,doi:10.1109/ICITACEE58587.2023.10277172.
6. P. Shrivastava, A. Kumari, S. Kumari and P. Bajaj, "A Comprehensive Review on the Prediction of Diabetes Disease Using Machine Learning," 2023 11th International Conference on Intelligent Systems and Embedded Design (ISED), Dehradun, India, 2023, pp. 1-6, doi: 10.1109/ISED59382.2023.10444546.
7. D. Ather, A. Singh, R. Kler, N. Chaudhary, B. Shukla and T. B. Z, "Application of Machine Learning in Predicting Diabetes: A Detailed Evaluation Using Support Vector Machine Classifier," 2024 1st International Conference on Sustainable Computing and Integrated Communication in Changing Landscape of AI (ICSCAI), Greater Noida, India, 2024, pp. 1-7, doi: 10.1109/ICSCAI61790.2024.10867028.
8. M. K. Daigavhane and S. Gundewar, "Diabetes Prediction using Different Machine Learning Classifiers," 2024 Parul International Conference on Engineering and Technology (PICET), Vadodara, India, 2024, pp. 1-6, doi: 10.1109/PICET60765.2024.10716110.
9. N. D'Souza, K. Shah and P. Singh, "Diabetes Detection Using Machine Learning Algorithms," 2022 IEEE Bombay Section Signature Conference (IBSSC), Mumbai, India, 2022, pp. 1-5, doi: 10.1109/IBSSC56953.2022.10037329.
10. D. R. Khadatkhar, A. Agrawal, A. Parveen and M. Kaif, "Diabetes Detection Using Deep Learning," 2023 International Conference on Artificial Intelligence for Innovations in Healthcare Industries (ICAIIHI), Raipur, India, 2023, pp. 1-6, doi: 10.1109/ICAIIHI57871.2023.10488978.
11. B. Badeji-Ajisafe et al., "Early Detection of Diabetes Using Supervised Learning Approach," 2024 International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG), Omu-Aran, Nigeria, 2024, pp. 1-8, doi: 10.1109/SEB4SDG60871.2024.10630330.
12. .A. H. Ataya, "Early detection of Diabetes using Machine Learning Techniques," 2023 Third International Conference on Artificial Intelligence and Smart Energy (ICAIS), Coimbatore, India, 2023, pp. 886-891, doi: 10.1109/ICAIS56108.2023.10073861.
13. S. Sharma and B. Sharma, "EDAS Based Selection of Machine Learning Algorithm for Diabetes Detection," 2020 9th International Conference System Modeling and Advancement in Research Trends (SMART), Moradabad, India, 2020, pp. 240-244, doi: 10.1109/SMART50582.2020.9337118.
14. A. C. Kulkarni and B. V. Satyasrikar, "Effects of Pre-Processing Techniques on Diabetes Detection Using Machine Learning," 2024 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC), Bhubaneswar, India, 2024, pp. 1-4, doi: 10.1109/ASSIC60049.2024.10508016.
15. N. K. Sinha, M. Pundir and P. Goel, "A Survey on Diabetes Mellitus Using Machine Learning Classifiers," 2025 2nd International Conference on Computational Intelligence, Communication Technology and Networking (CICTN), Ghaziabad, India, 2025, pp. 704-709, doi: 10.1109/CICTN64563.2025.10932388.

